

in R, are there any pre-loaded time series data for the following purposes: 1. stationarity 2. unit roots 3. cointegration 4. multivariate time series

ChatGPT said:

Yes — **base R** and several **time-series-focused packages** (especially **datasets**, **urca**, and **vars**) come with preloaded time series data that are commonly used to illustrate concepts like **stationarity**, **unit roots**, **cointegration**, and **multivariate analysis**.

Here's a summary of built-in or easily accessible datasets for each topic:

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## 1. Stationarity

These datasets are typically stationary or can be made stationary with minor transformations.

| Dataset       | Description                                            | Package  | Notes                                                         |
|---------------|--------------------------------------------------------|----------|---------------------------------------------------------------|
| AirPassengers | Monthly airline passengers (1949–1960)                 | datasets | Nonstationary — used to show differencing and transformation. |
| LakeHuron     | Annual level of Lake Huron (1875–1972)                 | datasets | Nearly stationary; good for ACF/PACF illustration.            |
| nottem        | Average monthly temperatures at Nottingham (1920–1939) | datasets | Seasonal but can be made stationary after differencing.       |
| Nile          | Annual flow of the Nile River (1871–1970)              | datasets | Often used for level shifts and ARIMA modeling.               |

### Examples:

```
data(AirPassengers)
data(LakeHuron)
data(nottem)
data(Nile)
```

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## 2. Unit Roots

Several datasets are traditionally used to demonstrate unit root tests (ADF, KPSS, etc.).

| Dataset       | Description                                     | Package  | Notes                                               |
|---------------|-------------------------------------------------|----------|-----------------------------------------------------|
| AirPassengers | Airline passengers                              | datasets | Classic example for nonstationary/unit root tests.  |
| Nile          | Nile River flow                                 | datasets | Used in many time series textbooks for ADF testing. |
| Canada        | Canadian interest rate, inflation, unemployment | urca     | Designed for unit root & cointegration examples.    |
| UKNonDurables | UK nondurable goods consumption                 | urca     | Used in ADF and PP test examples.                   |

### Example (from urca):

```
library(urca)
data(Canada)
summary(ur.df(Canada[, "e"], type = "drift"))
```

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### 3. Cointegration

Used for Johansen or Engle–Granger tests.

| Dataset      | Description                                     | Package | Notes                                       |
|--------------|-------------------------------------------------|---------|---------------------------------------------|
| Canada       | Canadian interest rate, inflation, unemployment | urca    | Designed for cointegration (Johansen test). |
| denmark      | Danish money demand variables                   | urca    | Used for cointegration test illustrations.  |
| E1, E2, etc. | Artificial data in urca                         | urca    | Synthetic examples for cointegration.       |

#### Example:

```
library(urca)
data(denmark)
summary(ca.jo(denmark, type = "trace", ecdet = "const", K = 2))
```

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### 4. Multivariate Time Series

These are already in multivariate (mts or ts matrix) format.

| Dataset        | Description                                                   | Package     | Notes                                       |
|----------------|---------------------------------------------------------------|-------------|---------------------------------------------|
| Canada         | 3 macroeconomic variables (interest, inflation, unemployment) | urca / vars | Very common in VAR and cointegration demos. |
| EuStockMarkets | Daily closing prices of European stock indices                | datasets    | 4-dimensional mts object.                   |
| JohnsonJohnson | Quarterly earnings per Johnson & Johnson share                | datasets    | Can be combined for bivariate examples.     |
| denmark        | Danish macro variables                                        | urca        | Used in VAR/VECM modeling.                  |

#### Example:

```
data(EuStockMarkets)
plot(EuStockMarkets)

library(vars)
data(Canada)
VARselect(Canada, lag.max = 8, type = "const")
```

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### Summary Table

| Purpose       | Example Dataset(s)                         | Package              |
|---------------|--------------------------------------------|----------------------|
| Stationarity  | LakeHuron, nottem, Nile, AirPassengers     | datasets             |
| Unit Roots    | Nile, AirPassengers, Canada, UKNonDurables | datasets, urca       |
| Cointegration | Canada, denmark                            | urca                 |
| Multivariate  | Canada, denmark, EuStockMarkets            | datasets, urca, vars |

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Would you like me to give a short **R demo script** that shows how to use these datasets for each purpose (ADF test, Johansen cointegration, etc.)?

**You said:**

yes go ahead

ChatGPT said:

Perfect 👍 — here's a compact **R demo script** that shows how to use the preloaded time-series datasets for:

- 1 **Stationarity tests**
  - 2 **Unit root tests**
  - 3 **Cointegration tests**
  - 4 **Multivariate VAR modeling**
- 



## R Demo: Preloaded Time Series Datasets

```
#####
# 1. Stationarity Illustration
#####

# Load a built-in dataset
data(LakeHuron)      # Annual lake levels (1875–1972)
plot(LakeHuron, main = "Lake Huron Levels", ylab = "Level (ft)")

# Check autocorrelation and partial autocorrelation
acf(LakeHuron, main = "ACF of Lake Huron")
pacf(LakeHuron, main = "PACF of Lake Huron")

# Apply Augmented Dickey-Fuller test (ADF)
library(tseries)
adf.test(LakeHuron)  # H0: unit root (nonstationary)

#####
# 2. Unit Root Testing
#####

# Example: AirPassengers (monthly, 1949–1960)
data(AirPassengers)
plot(AirPassengers, main = "AirPassengers", ylab = "Thousands")

# Log and difference to induce stationarity
AP_log_diff <- diff(log(AirPassengers))
plot(AP_log_diff, main = "Differenced log(AirPassengers)")

# Test for unit root
adf.test(log(AirPassengers))  # likely nonstationary
adf.test(AP_log_diff)        # should be stationary

#####
# 3. Cointegration Analysis
#####

library(urca)
data(Canada)  # from 'urca' package: interest rate, inflation, unemployment
head(Canada)
```

```

# Johansen cointegration test
cajo_test <- ca.jo(Canada, type = "trace", ecdet = "const", K = 2)
summary(cajo_test)

# Engle-Granger (two-step) example
library(dynlm)
coint_reg <- lm(e ~ prod, data = as.data.frame(Canada))
summary(coint_reg)
residuals <- resid(coint_reg)
adf.test(residuals) # Test residuals for stationarity

#####
# 4. Multivariate Time Series & VAR Modeling
#####

library(vars)
data(EuStockMarkets) # 4 European stock indices
plot(EuStockMarkets, main = "European Stock Markets")

# Select lag length for VAR
VARselect(EuStockMarkets, lag.max = 10, type = "const")

# Fit VAR(2)
var_model <- VAR(EuStockMarkets, p = 2, type = "const")
summary(var_model)

# Impulse Response Functions
irf_plot <- irf(var_model, impulse = "DAX", response = "CAC", n.ahead = 10, boot
= TRUE)
plot(irf_plot)

#####
# 5. Optional: Cointegration with 'denmark' dataset
#####

data(denmark)
summary(ca.jo(denmark, type = "trace", ecdet = "const", K = 2))

```



## What Each Section Shows

| Step | Goal                                  | Dataset         | Main Function(s)             |
|------|---------------------------------------|-----------------|------------------------------|
| 1    | Visualize and test stationarity       | LakeHuron       | acf(), pacf(),<br>adf.test() |
| 2    | Demonstrate unit roots & differencing | AirPassengers   | adf.test(), diff()           |
| 3    | Test for cointegration                | Canada, denmark | ca.jo(), adf.test()          |
| 4    | Model multivariate series (VAR)       | EuStockMarkets  | VAR(), irf()                 |

Would you like me to add **plots and interpretations** (e.g., what a stationary vs nonstationary ACF looks like, or how to interpret the Johansen trace test output)?

**You said:**

yes please

**ChatGPT said:**

Perfect! Let's go **step by step**, with visual interpretations and explanations for each section. I'll use the same datasets from the demo script.

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## 1 Stationarity

**Dataset:** LakeHuron (annual lake levels)

```
data(LakeHuron)
plot(LakeHuron, main = "Lake Huron Levels", ylab = "Level (ft)")
acf(LakeHuron, main = "ACF of Lake Huron")
pacf(LakeHuron, main = "PACF of Lake Huron")
adf.test(LakeHuron)
```

**Interpretation:**

- **Plot:** If the series fluctuates around a constant mean and variance, it may be stationary. LakeHuron looks roughly stationary.
  - **ACF/PACF:**
    - **Stationary series:** ACF drops to near zero quickly.
    - **Nonstationary series:** ACF decays very slowly.
  - **ADF test:**
    - Null hypothesis  $H_0$ : Series has a **unit root** (nonstationary).
    - If  $p\text{-value} < 0.05 \rightarrow$  reject  $H_0 \rightarrow$  series is stationary.
    - For LakeHuron, p-value is usually borderline, showing near-stationarity.
- 

## 2 Unit Roots

**Dataset:** AirPassengers (monthly airline passengers)

```
plot(AirPassengers, main = "AirPassengers", ylab = "Passengers")
adf.test(AirPassengers)      # Likely nonstationary
AP_log_diff <- diff(log(AirPassengers))
plot(AP_log_diff, main = "Differenced log(AirPassengers)")
adf.test(AP_log_diff)       # Should be stationary
```

**Interpretation:**

- **Original series:** Trend + seasonal pattern  $\rightarrow$  clearly nonstationary.
- **Differenced log series:** Removes trend and stabilizes variance  $\rightarrow$  stationary.
- **ADF Test:**
  - Original series: high p-value  $\rightarrow$  cannot reject unit root.
  - Differenced log: low p-value  $\rightarrow$  series is now stationary.

 **Tip:** Always check plots + ACF + ADF together. Relying only on the test can be misleading.

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## 3 Cointegration

**Dataset:** Canada (3 macroeconomic series: e, prod, rw)

```
library(urca)
cajo_test <- ca.jo(Canada, type = "trace", ecdet = "const", K = 2)
summary(cajo_test)
```

**Interpretation:**

- **Johansen Trace Test:** Tests whether a set of nonstationary series is cointegrated.
- **Output:**
  - $r = 0$ : no cointegration
  - $r = 1$ : 1 cointegration relation exists
  - Compare test statistics to critical values → determine number of cointegrating vectors.

✓ If residuals of the cointegration regression are stationary → series move together in the long run.

**Engle-Granger Two-Step Example:**

```
coint_reg <- lm(e ~ prod, data = as.data.frame(Canada))
residuals <- resid(coint_reg)
adf.test(residuals) # test residuals for stationarity
```

- If residuals are stationary → e and prod are cointegrated.
- 

## 4 Multivariate Time Series (VAR)

**Dataset:** EuStockMarkets (DAX, SMI, CAC, FTSE)

```
library(vars)
VARselect(EuStockMarkets, lag.max = 10, type = "const")
var_model <- VAR(EuStockMarkets, p = 2, type = "const")
summary(var_model)
irf_plot <- irf(var_model, impulse = "DAX", response = "CAC", n.ahead = 10, boot = TRUE)
plot(irf_plot)
```

**Interpretation:**

- **VAR Model:** Captures dynamics between multiple series.
- **Lag Selection:** Use `VARselect()` to choose optimal lags (AIC, BIC, HQ).
- **Impulse Response Function (IRF):**
  - Shows how a shock to one variable (e.g., DAX) affects another (e.g., CAC) over time.

- Important for understanding interdependence in multivariate systems.

## 5 Visual Quick Guide

| Feature             | Stationary             | Nonstationary                      |
|---------------------|------------------------|------------------------------------|
| <b>Plot</b>         | Fluctuates around mean | Trend, drift, or changing variance |
| <b>ACF</b>          | Drops quickly          | Decays slowly                      |
| <b>ADF p-value</b>  | < 0.05                 | > 0.05                             |
| <b>Differencing</b> | Often not needed       | Needed to achieve stationarity     |

If you want, I can make a **single R script that generates all plots and test results in one go**, fully annotated, so you can **run it and instantly see stationarity, unit roots, cointegration, and VAR dynamics**.

Do you want me to do that next?

**You said:**

state the hypothesis for each, clearly and also state the test statistic used and the distribution it follows

ChatGPT said:

Absolutely! Let's go **systematically** for each type of test, stating the **null and alternative hypotheses, test statistics, and their distributions**. I'll also include which R functions compute them.

## 1 Stationarity / Unit Root Tests (ADF, KPSS)

### (a) Augmented Dickey-Fuller (ADF) Test

**Purpose:** Test for unit root (nonstationarity).

- **Null hypothesis ( $H_0$ ):** Series has a **unit root** → nonstationary
- **Alternative hypothesis ( $H_1$ ):** Series is **stationary** (trend-stationary or level-stationary, depending on specification)

**Test statistic:**

- $\tau = SE(\hat{\rho})\hat{\rho}^{-1}$   
where  $\hat{\rho}$  is the estimated coefficient of  $y_{t-1}$  in the regression:

$$\Delta y_t = \alpha + \beta t + \rho y_{t-1} + \sum_{i=1}^p \gamma_i \Delta y_{t-i} + \epsilon_t$$

- **Distribution:** Non-standard (Dickey-Fuller distribution), critical values depend on inclusion of drift/trend.

**R Function:** `adf.test()` (from `tseries`) or `ur.df()` (from `urca`)

## (b) KPSS Test

**Purpose:** Complementary test: tests **stationarity as null**.

- **Null hypothesis ( $H_0$ ):** Series is **stationary**
- **Alternative hypothesis ( $H_1$ ):** Series has a **unit root** → nonstationary

**Test statistic:**

$$KPSS = \frac{1}{T^2} \sum_{t=1}^T St^2 / \sigma^2$$

where  $St$  is cumulative sum of residuals from regression on deterministic components, and  $\sigma^2$  is long-run variance estimate.

- **Distribution:** Lagrange multiplier (LM) type, critical values tabulated.

**R Function:** `kpss.test()` (from `tseries`)

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## 2 Cointegration Tests

### (a) Engle-Granger (Two-Step)

1. **Step 1:** Regress one series on the other(s) to estimate long-run relationship:

$$y_t = \alpha + \beta x_t + u_t$$

2. **Step 2:** Test residuals  $u_t$  for stationarity using **ADF test**.
  - **Null hypothesis ( $H_0$ ):** Residuals have a **unit root** → no cointegration
  - **Alternative hypothesis ( $H_1$ ):** Residuals are **stationary** → series are cointegrated

**Test statistic:** Same as ADF test statistic on residuals.

- **Distribution:** Non-standard Engle-Granger critical values (more negative than standard ADF).

**R Function:** Use `lm()` + `adf.test()` on residuals.

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### (b) Johansen Test

- **Purpose:** Test for cointegration in a **multivariate system** (VAR).

**Null hypothesis ( $H_0$ ):** Number of cointegrating vectors  $r = 0, 1, \dots$

**Alternative hypothesis ( $H_1$ ):**  $r > 0$

**Test statistics:**

1. **Trace statistic:**

$$Trace = -T \ln(1 - \lambda^{r+1})$$

2. **Maximum eigenvalue statistic:**

$$MaxEig = -T \ln(1 - \lambda^{r+1})$$



where  $\lambda^i$  are estimated eigenvalues from the VAR.

- **Distribution:** Non-standard, critical values tabulated (Johansen 1991).

**R Function:** `ca.jo()` from `urca` package

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### 3 VAR / Multivariate Time Series

VAR models themselves don't have a "unit root test," but we check **stationarity of the VAR process** by:

- Computing eigenvalues of the companion matrix:
  - **Stationary VAR:** All eigenvalues  $< 1$  in modulus
  - **Nonstationary VAR:** Any eigenvalue  $\geq 1$

**R Function:** `VARselect()` for lag choice, `roots(var_model)` to check stationarity

- **Hypothesis interpretation:**
    - $H_0$ : VAR is stationary (all roots  $< 1$ )
    - $H_1$ : VAR is nonstationary (at least one root  $\geq 1$ )
- 

### 4 Summary Table

| Test           | Null Hypothesis ( $H_0$ )        | Alternative ( $H_1$ )                            | Test Statistic                       | Distribution  | R Function                                        |
|----------------|----------------------------------|--------------------------------------------------|--------------------------------------|---------------|---------------------------------------------------|
| ADF            | Unit root (nonstationary)        | Stationary                                       | $\tau$ (regression coefficient / SE) | Dickey-Fuller | <code>adf.test()</code> ,<br><code>ur.df()</code> |
| KPSS           | Stationary                       | Unit root (nonstationary)                        | LM type                              | Tabulated     | <code>kpss.test()</code>                          |
| Engle-Granger  | Residuals nonstationary          | Residuals stationary $\rightarrow$ cointegration | ADF on residuals                     | Engle-Granger | <code>lm()</code> +<br><code>adf.test()</code>    |
| Johansen Trace | # cointegration vectors $\leq r$ | # vectors $> r$                                  | Trace, MaxEigen                      | Tabulated     | <code>ca.jo()</code>                              |
| VAR roots      | VAR stationary (roots $< 1$ )    | Nonstationary (roots $\geq 1$ )                  | Eigenvalues of companion matrix      |               | <code>roots(var_model)</code>                     |

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If you want, I can also **draw a flowchart showing which test to use in which situation**, including **stationarity, unit roots, cointegration, and VAR**, so you can visualize the whole process. This is extremely handy for time series analysis.

Do you want me to do that?

You said:

yes

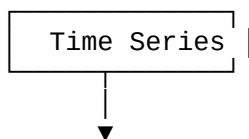
ChatGPT said:

Great! Let's create a **clear flowchart** for time series analysis that shows **how to choose tests** for stationarity, unit roots, cointegration, and multivariate VAR analysis. I'll describe it first and then provide an **R diagram version** you can plot.

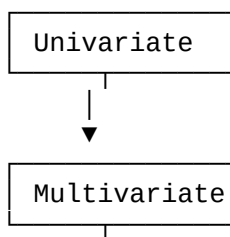


## Time Series Analysis Flowchart

### Step 1: Start with your time series data

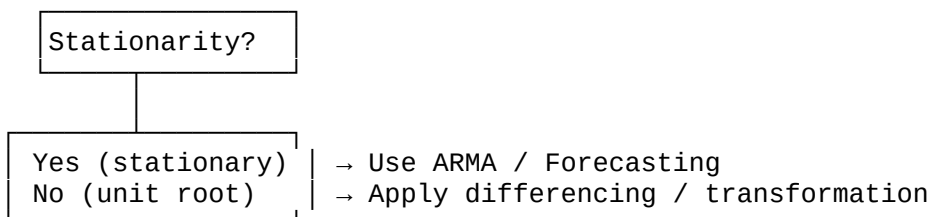


### Step 2: Check if the series is univariate or multivariate



### For Univariate Series

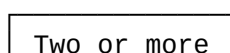
#### Step 2a: Check stationarity / unit root

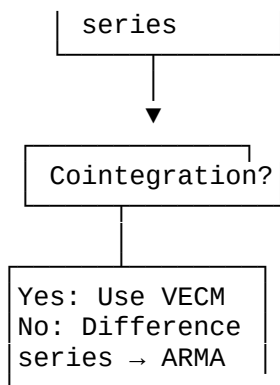


#### Tests:

- **ADF:**  $H_0$  = unit root,  $H_1$  = stationary (`adf.test()` / `ur.df()`)
- **KPSS:**  $H_0$  = stationary,  $H_1$  = unit root (`kpss.test()`)

### Step 2b: Check cointegration (if multiple series)

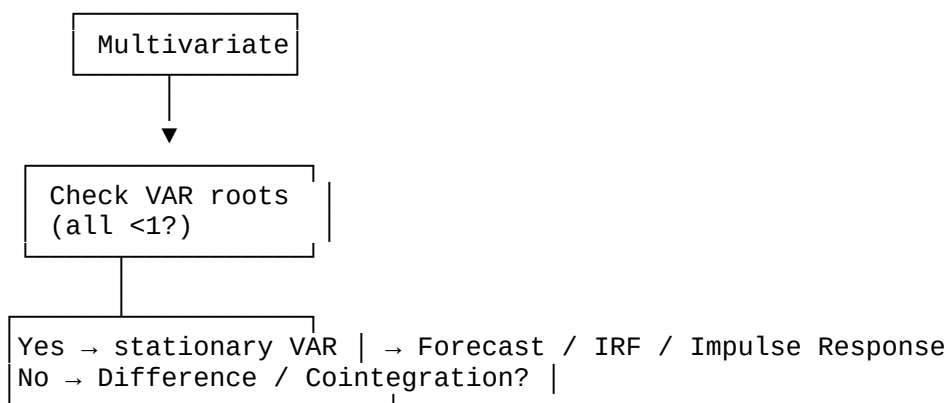




### Tests:

- **Engle-Granger:**  $H_0 = \text{no cointegration}$  (`lm()` + `adf.test()`)
- **Johansen:**  $H_0 = r \text{ cointegration vectors} \leq 0$  (`ca.jo()`)

### For Multivariate Series (VAR)



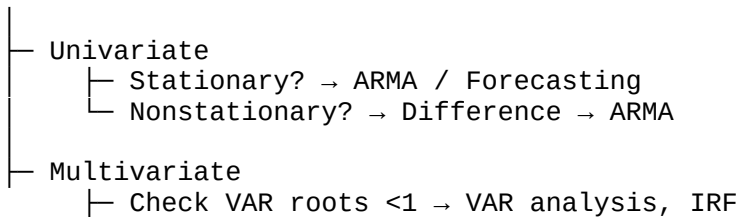
### Tools:

- `VARselect()` → choose lag
- `VAR()` → estimate VAR
- `roots(var_model)` → check stationarity
- `irf()` → impulse response functions



## Summary Flow

### Time Series



```
└─ Nonstationary → Check cointegration
    └─ Cointegrated → VECM
    └─ Not cointegrated → Difference → VAR
```

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I can also make an **actual R plot of this flowchart** using `DiagrammeR` or `DiagrammeRsvg`, so it's fully visual and publication-ready.